



# CSE 445

## Lecture 7

Linear classifier by decision boundaries

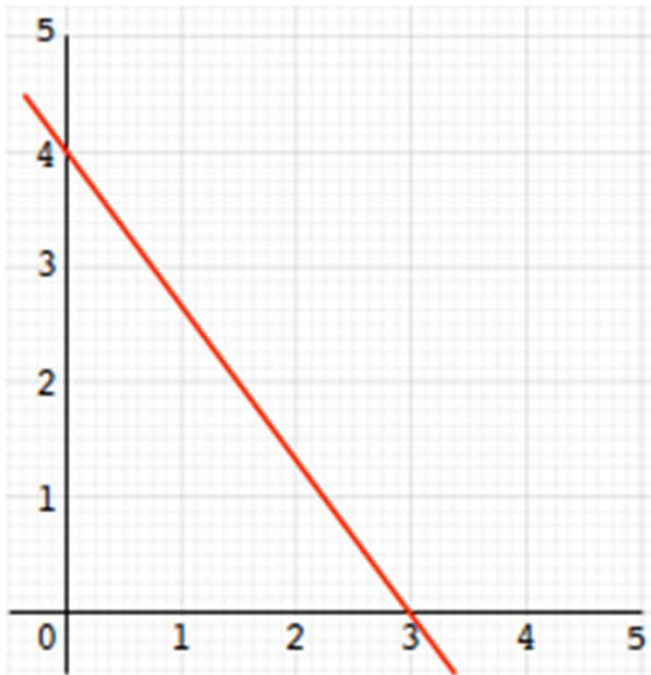


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# Linear Classifier

- Linear decision boundary for binary classification
- A loss function for classification
- The perceptron algorithm

# Linear decision boundary for classification



- What is the formula for this boundary?
- What label would we predict for a new point  $x$ ?

# Linear classifiers

**Binary classification problem: data  $x \in \mathbb{R}^d$  and labels  $y \in \{-1, +1\}$**

- Linear classifier:
  - Parameters:  $w \in \mathbb{R}^d$  and  $b \in \mathbb{R}$
  - Decision boundary  $w \cdot x + b = 0$
  - On point  $x$ , predict label  $\text{sign}(w \cdot x + b)$
- If the true label on point  $x$  is  $y$ :
  - Classifier correct if  $y(w \cdot x + b) > 0$

# A loss function for classification

What is the **loss** of our linear classifier (given by  $w, b$ ) on a point  $(x, y)$ ?

One idea for a loss function:

- If  $y(w \cdot x + b) > 0$ : correct, no loss
- If  $y(w \cdot x + b) < 0$ :  $\text{loss} = -y(w \cdot x + b)$

# A simple learning algorithm

Fit a linear classifier  $w, b$  to the training set using **stochastic gradient descent**.

- Update  $w, b$  based on just one data point  $(x, y)$  at a time
- If  $y(w \cdot x + b) > 0$ : zero loss, no update
- If  $y(w \cdot x + b) \leq 0$ : loss is  $-y(w \cdot x + b)$

# A simple learning algorithm

Fit a linear classifier  $w, b$  to the training set using **stochastic gradient descent**.

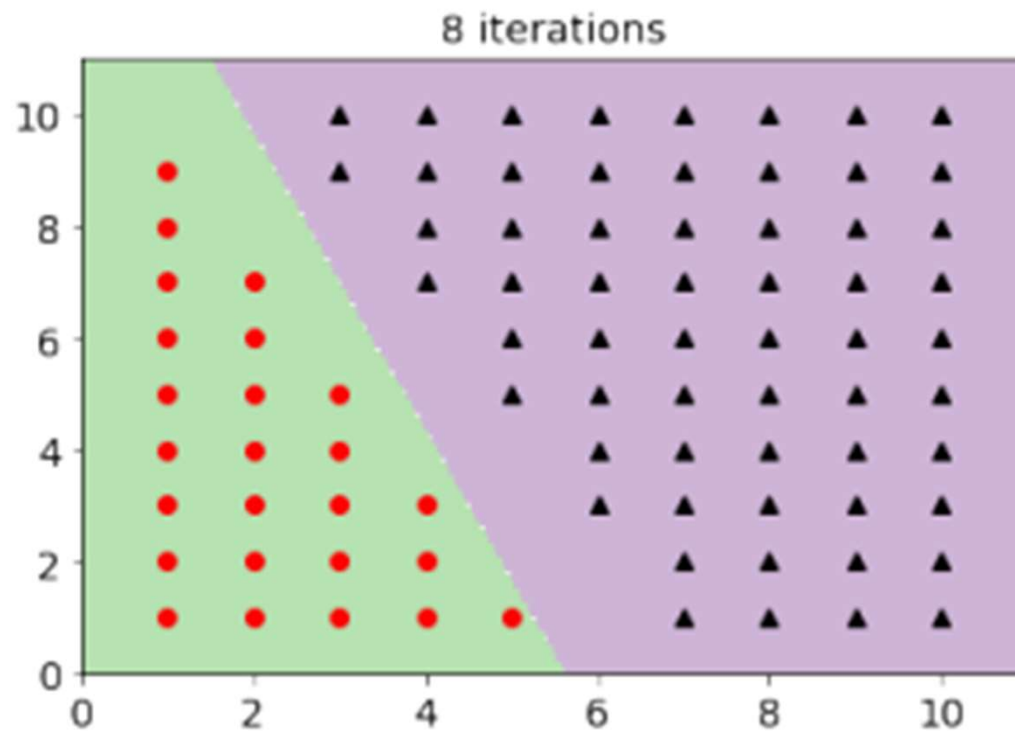
- Update  $w, b$  based on just one data point  $(x, y)$  at a time
- If  $y(w \cdot x + b) > 0$ : zero loss, no update
- If  $y(w \cdot x + b) \leq 0$ : loss is  $-y(w \cdot x + b)$

## The Perceptron algorithm

- Initialize  $w = 0$  and  $b = 0$
- Keep cycling through the training data  $(x, y)$ :
  - If  $y(w \cdot x + b) \leq 0$  (i.e. point misclassified):
    - $w = w + yx$
    - $b = b + y$

# Perceptron algorithm example

85 data points – linearly separable

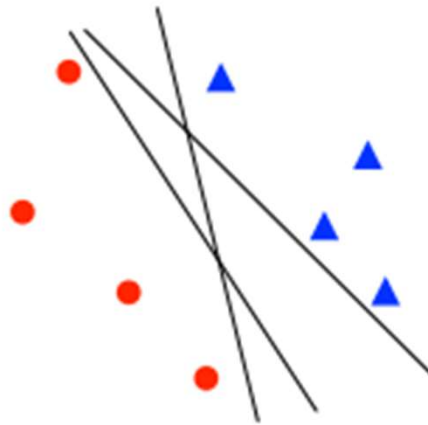




# Perceptron convergence

If the training data is linearly separable:

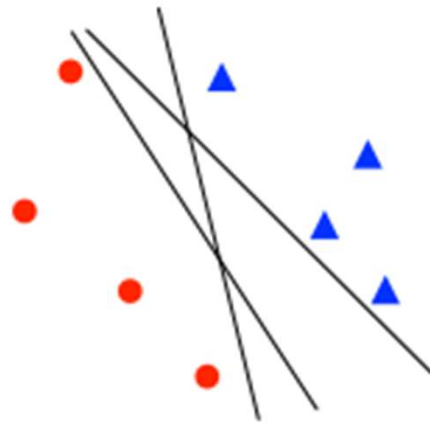
- The Perceptron algorithm will find a linear classifier with zero training error
- It will converge within a finite number of steps.



But is there a better, more systematic choice of separator?

# Improving the perceptron

For a linearly separable data set, there are in general many possible separating hyperplanes, and Perceptron is guaranteed to find one of them.



Is there a better, more systematic choice of separator?  
The one with the most buffer around it, for instance?

# The learning problem

Given: training data  $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathbb{R}^d \times \{-1, +1\}$ .

Find:  $w \in \mathbb{R}^d$  and  $b \in \mathbb{R}$  such that  $y^{(i)}(w \cdot x^{(i)} + b) > 0$  for all  $i$ .

By scaling  $w, b$ , can equivalently ask for

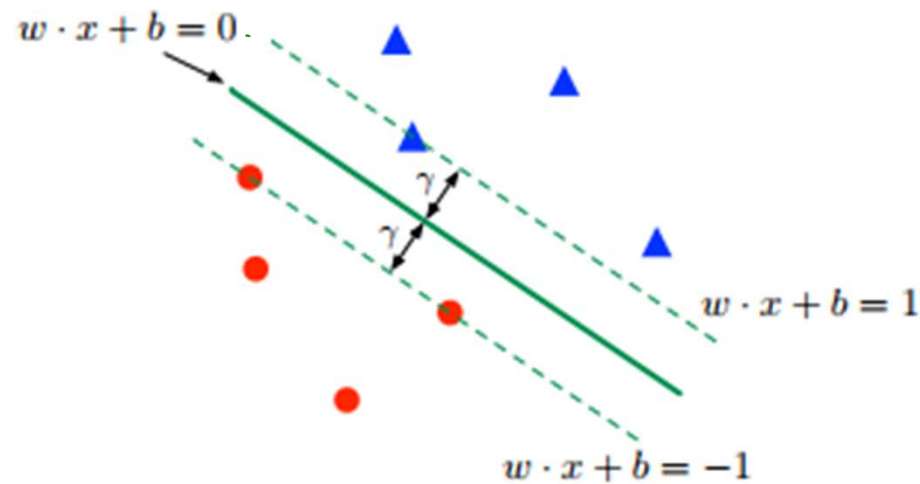
$$y^{(i)}(w \cdot x^{(i)} + b) \geq 1 \quad \text{for all } i$$

# Maximizing the margin

Given: training data  $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathbb{R}^d \times \{-1, +1\}$ .

Find:  $w \in \mathbb{R}^d$  and  $b \in \mathbb{R}$  such that

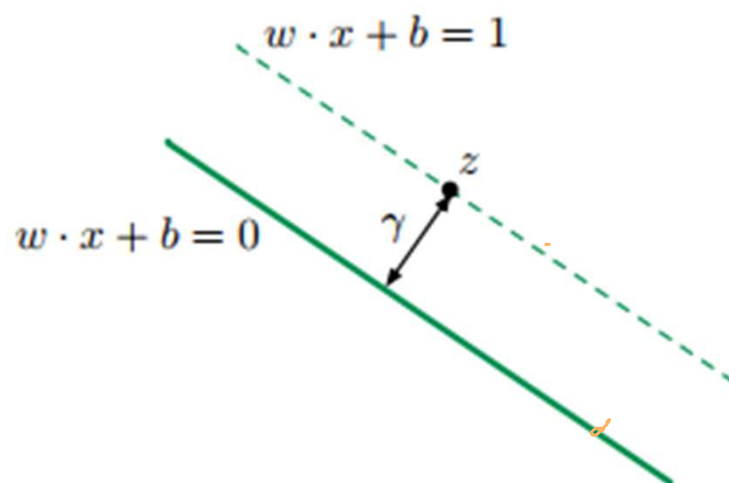
$$y^{(i)}(w \cdot x^{(i)} + b) \geq 1 \quad \text{for all } i.$$



Maximize the **margin**  $\gamma$ .

# Formula for a margin

Close-up of a point  $z$  on the positive boundary.



A quick calculation shows that  $\gamma = 1/\|w\|$ .

In short: to maximize the margin, minimize  $\|w\|$ .

# Maximum margin linear classifier

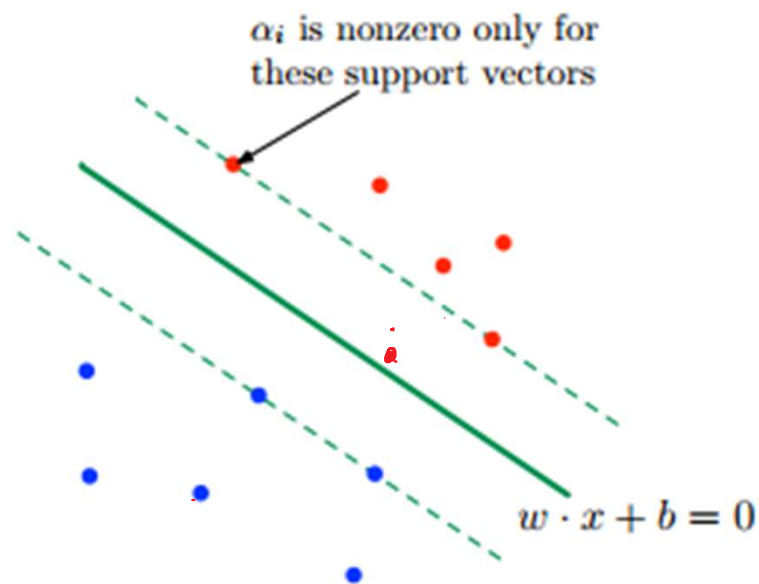
- Given  $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathbb{R}^d \times \{-1, +1\}$

$$\begin{aligned} \min_{w \in \mathbb{R}^d, b \in \mathbb{R}} \quad & \|w\|^2 \\ \text{s.t.:} \quad & y^{(i)}(w \cdot x^{(i)} + b) \geq 1 \quad \text{for all } i = 1, 2, \dots, n \end{aligned}$$

- This is a **convex optimization problem**:
  - Convex objective function
  - Linear constraints
- This means that:
  - the optimal solution can be found efficiently
  - duality** gives us information about the solution

# Support vectors

**Support vectors:** training points right on the margin, i.e.  $y^{(i)}(w \cdot x^{(i)} + b) = 1$ .



$w = \sum_{i=1}^n \alpha_i y^{(i)} x^{(i)}$  is a function of just the support vectors.

# SVM example: Iris data set

Fisher's **iris** data



150 data points from three classes:

- iris setosa
- iris versicolor
- iris virginica

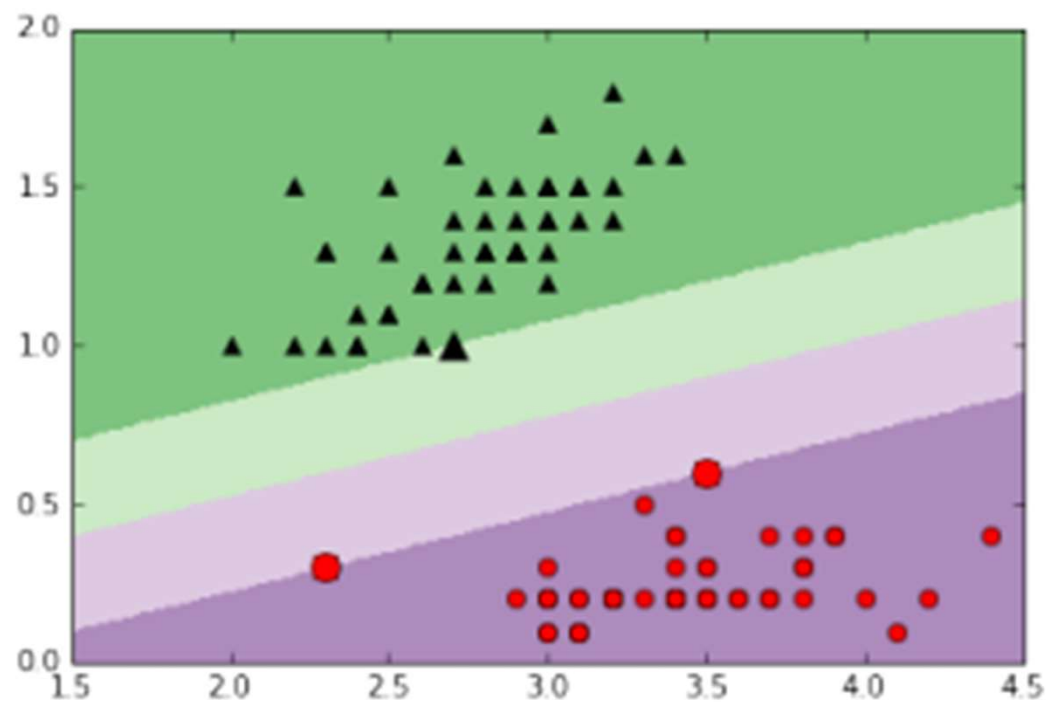
Four measurements: petal width/length, sepal width/length



# SVM example: Iris data set

Two features: sepal width, petal width.

Two classes: setosa (red circles), versicolor (black triangles)

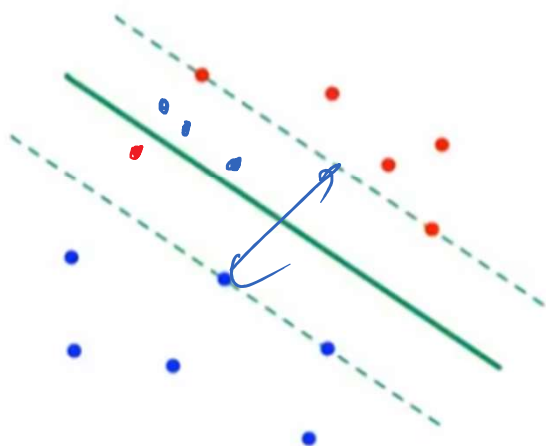


# Maximum margin classifier

Given:  $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathbb{R}^d \times \{-1, +1\}$ .

Find: the linear separator  $w$  that perfectly classifies the data and has maximum margin.

$$\begin{aligned} \min_{w \in \mathbb{R}^d, b \in \mathbb{R}} \quad & \|w\|^2 \\ \text{s.t.:} \quad & y^{(i)}(w \cdot x^{(i)} + b) \geq 1 \quad \text{for all } i = 1, 2, \dots, n \end{aligned}$$

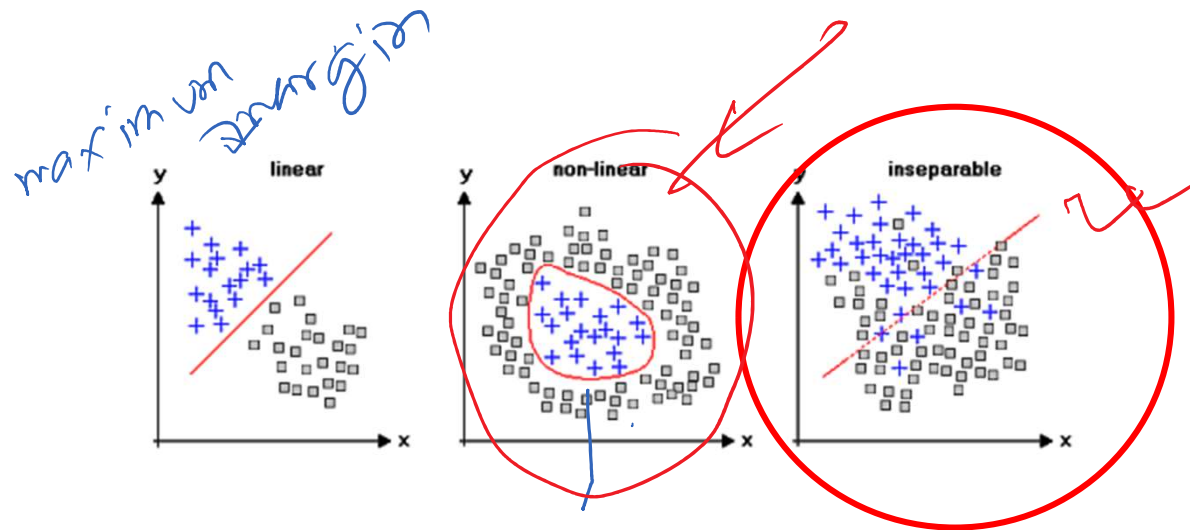


Solution  $w = \sum_{i=1}^n \alpha_i y^{(i)} x^{(i)}$  is a function of just the support vectors.

**What if data is not separable?**

# Soft margin SVM

- Data that isn't linearly separable
- The idea is to add slack variables for each data point
- We need to reformulate the optimization problem
- We also set the slack parameters



# Soft margin classifier

Idea: allow each data point  $x^{(i)}$  some slack  $\xi_i$ .

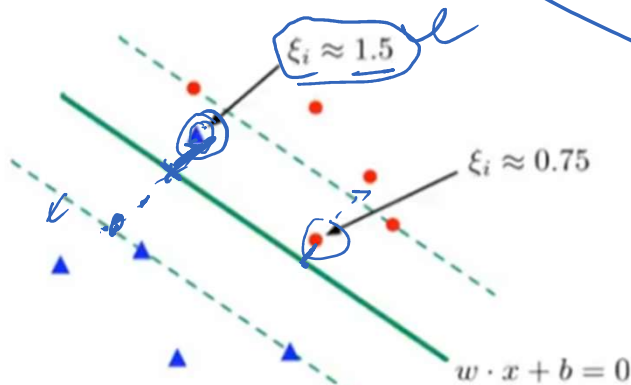
$$\min_{w \in \mathbb{R}^d, b \in \mathbb{R}, \xi \in \mathbb{R}^n} \|w\|^2 + C \sum_{i=1}^n \xi_i$$

$$\text{s.t.: } y^{(i)}(w \cdot x^{(i)} + b) \geq 1 - \xi_i \text{ for all } i = 1, 2, \dots, n$$

$$\xi_i \geq 0$$

$$wx + b = -1$$

$$1.5 + 0.75 = 2.25$$



$$y(wx + b) = 1 - 1.5 = -0.5$$

$$y(wx + b) = 1 - 0.75 = 0.25$$

$$\begin{pmatrix} x^{(1)} & y^{(1)} \\ x^{(2)} & y^{(2)} \end{pmatrix}$$

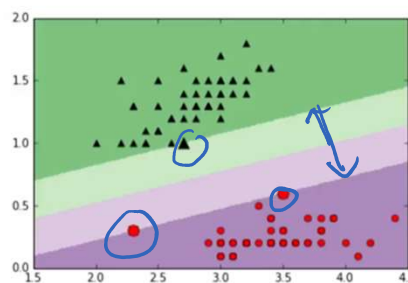
$$C = 0$$

$$C = 0$$

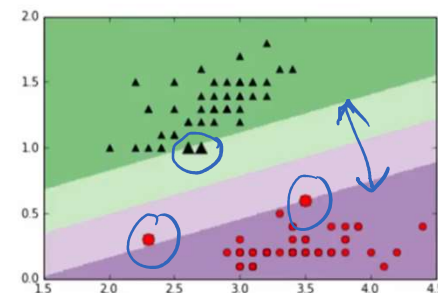
# Margin vs Slack

$$\begin{aligned} \min_{w \in \mathbb{R}^d, b \in \mathbb{R}, \xi \in \mathbb{R}^n} \quad & \|w\|^2 + C \sum_{i=1}^n \xi_i \\ \text{s.t.:} \quad & y^{(i)}(w \cdot x^{(i)} + b) \geq 1 - \xi_i \quad \text{for all } i = 1, 2, \dots, n \\ & \xi \geq 0 \end{aligned}$$

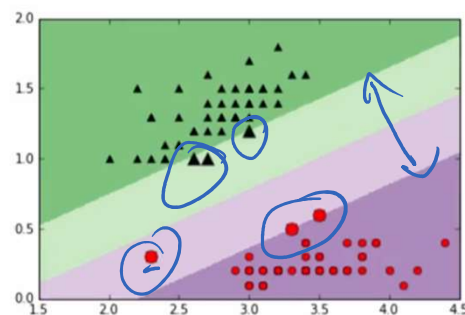
$C = 10$



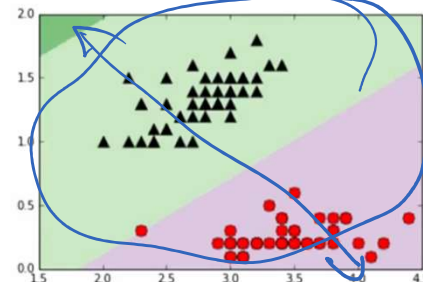
$C = 3$



$C = 2$



$C = 0.01$



# Non-linear boundary

- We can use these linear models to do even more than separating
  - Linearly separable classes and
  - Inseparable classes
- We can find separating boundaries that at
  - ✓ • Quadratic or even
  - ✓ • More degrees
- We need to understand the concept of duality for doing this efficiently

# Duality

Given a training set of points  $\{(x^{(i)}, y^{(i)}) : i = 1 \dots n\}$ :

## Perceptron algorithm

- Initialize  $w = 0$  and  $b = 0$
- While some training point  $(x, y)$  is misclassified:
  - $w = w + yx$
  - $b = b + y$

The final answer is of the form:

$$w = \sum_i \alpha_i y^{(i)} x^{(i)},$$

where  $\alpha_i = \#$  of times an update occurred on point  $i$ .

Can equivalently represent  $w$  by  $\alpha = (\alpha_1, \dots, \alpha_n)$ .

# Dual form of the perceptron algorithm

## Perceptron algorithm: primal form

- Initialize  $w = 0$  and  $b = 0$
- While some training point  $(x^{(i)}, y^{(i)})$  is misclassified:
  - $w = w + y^{(i)}x^{(i)}$
  - $b = b + y^{(i)}$

## Perceptron algorithm: dual form

- Initialize  $\alpha = 0$  and  $b = 0$
- While some training point  $(x^{(i)}, y^{(i)})$  is misclassified:
  - $\alpha_i = \alpha_i + 1$
  - $b = b + y^{(i)}$

Answer:  $w = \sum_i \alpha_i y^{(i)} x^{(i)}$



# Hard margin SVM

- Given  $(x^{(1)}, y^{(1)}), \dots, (x^{(n)}, y^{(n)}) \in \mathbb{R}^d \times \{-1, +1\}$

$$\begin{array}{ll} \text{(PRIMAL)} & \min_{w \in \mathbb{R}^d, b \in \mathbb{R}} \|w\|^2 \\ \text{s.t.:} & y^{(i)}(w \cdot x^{(i)} + b) \geq 1 \quad \text{for all } i = 1, 2, \dots, n \end{array}$$

- This is a **convex optimization problem**:
  - Convex objective function
  - Linear constraints
  - As such, it has a **dual maximization problem**.
  - The **primal** and **dual** problems have the same optimum value.

# Hard margin dual problem

$$\begin{array}{ll} \text{(PRIMAL)} & \min_{w \in \mathbb{R}^d, b \in \mathbb{R}} \|w\|^2 \\ \text{s.t.:} & y^{(i)}(w \cdot x^{(i)} + b) \geq 1 \quad \text{for all } i = 1, 2, \dots, n \end{array}$$

$$\begin{array}{ll} \text{(DUAL)} & \max_{\alpha \in \mathbb{R}^n} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} (x^{(i)} \cdot x^{(j)}) \\ & \text{s.t.:} \quad \sum_{i=1}^n \alpha_i y^{(i)} = 0 \\ & \quad \alpha \geq 0 \end{array}$$

Complementary slackness: At optimality,  $w = \sum_{i=1}^n \alpha_i y^{(i)} x^{(i)}$  and

$$\alpha_i > 0 \Rightarrow y^{(i)}(w \cdot x^{(i)} + b) = 1$$

Points  $x^{(i)}$  with  $\alpha_i > 0$  are **support vectors**.

# Dual of soft-margin SVM

$$\begin{aligned}
 & \text{(PRIMAL)} \quad \min_{w \in \mathbb{R}^d, b \in \mathbb{R}, \xi \in \mathbb{R}^n} \quad \|w\|^2 + C \sum_{i=1}^n \xi_i \\
 & \text{s.t.:} \quad y^{(i)}(w \cdot x^{(i)} + b) \geq 1 - \xi_i \quad \text{for all } i = 1, 2, \dots, n \\
 & \quad \quad \quad \xi \geq 0
 \end{aligned}$$

$$\begin{aligned}
 & \text{(DUAL)} \quad \max_{\alpha \in \mathbb{R}^n} \quad \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y^{(i)} y^{(j)} (x^{(i)} \cdot x^{(j)}) \\
 & \quad \quad \quad \text{s.t.:} \quad \sum_{i=1}^n \alpha_i y^{(i)} = 0 \\
 & \quad \quad \quad 0 \leq \alpha_i \leq C
 \end{aligned}$$

At optimality,  $w = \sum_i \alpha_i y^{(i)} x^{(i)}$ , with

$$0 < \alpha_i < C \Rightarrow y^{(i)}(w \cdot x^{(i)} + b) = 1$$

$$\alpha_i = C \Rightarrow y^{(i)}(w \cdot x^{(i)} + b) = 1 - \xi_i$$